8. THE PRACTICAL APPLICATIONS OF NUCLEAR PHYSICS

I. THE EXPLOITATION OF ATOMIC ENERGY FOR USEFUL PURPOSES

In studying the practical applications of nuclear physics, it is helpful to start out with analogies with chemistry. Chemistry deals with the combination of different elements in more complex substances, the chemical compounds, or conversely, with the separation of elements from such compounds, whereas nuclear physics deals with the transmutation of one element into another. Chemical processes serve two fundamentally different purposes: First, they can be used to convert less valuable materials into others of greater value, as, for instance, the combination of carbon and hydrogen to form benzol; secondly, a chemical conversion can be utilized in order to obtain energy, as, for instance, the combustion of coal to form carbon dioxide, in order to produce heat. It is obvious that these two applications are not independent of each other. It is often the case that a material is produced solely in order to be used as a source of energy—as, for instance, benzol.

The same is true, we may say, of nuclear processes. They can be applied, first, to produce a more valuable material out of one of lesser value, and secondly, they can be instrumental in producing energy.

In order to gain an idea of the order of magnitude of the energy which can be obtained through nuclear reactions, let us make use of another analogy with chemistry. The combustion of carbon with oxygen resulting in the formation of carbon dioxide can be written in the form of an equation which also shows the energy yielded by this process:

$$C + O_2 \rightarrow CO_2 + 96 \text{ kilocalories}$$

This formula relates to units of 1 mole and 1 gramme-atom, and states that the combustion of 1 gramme-atom = 12 grammes of carbon with 1 mole = 32 grammes of gaseous oxygen produces 1 mole = 44 grammes of carbon dioxide and liberates 96 kilocalories of heat.

Another example is the combustion of oxygen with hydrogen, resulting in